

Jinu Lee

Email ◇ Homepage ◇ GitHub ◇ LinkedIn

RESEARCH INTERESTS

My research focuses on **understanding reasoning traces of large language models**. Specific topics include:

- Understanding the text-level and mechanistic-level structures of LLM reasoning
- Evaluating and improving the reasoning capability of LLMs
- Trustworthy reasoning in high-stakes domains (e.g., law, tax, medicine, ...)

EDUCATION

University of Illinois Urbana-Champaign Ph.D., Computer Science / Adviser: Dr. Julia Hockenmaier	Urbana-Champaign, IL 08/2024 - Present
Seoul National University B.S., Computer Science & Engineering	Seoul, Rep. of Korea 03/2018 - 08/2024

WORK EXPERIENCE

Microsoft Research Research Intern	Redmond, WA 05/2025 - 08/2025, 05/2026 - 08/2026
• Post-training LLM for program verification and theorem proving (2025) • Autoformalizing tool-calling environment constraints and user intent (2026)	
LBOX Research Engineer	Seoul, Rep. of Korea 07/2023 - 06/2024
• Neuro-symbolic reasoning; legal information extraction	
NCSoft Researcher	Seongnam, Rep. of Korea 07/2020 - 11/2020
• Korean dependency parsers and resources	

PUBLICATIONS

Conference proceedings.

- [C8] **Jinu Lee**, Shivam Agarwal, Amruta Parulekar, Siddarth Madala, Dilek Hakkani-Tur, Julia Hockenmaier (2026). ReasoningFlow: Discourse Structures for Understanding LLM Reasoning Traces. *EMNLP 2026 Main Conference (Oral)*.
- [C7] **Jinu Lee**, Kyoung-woon On, Simeng Han, Arman Cohan, Julia Hockenmaier (2026). Evaluating Legal Reasoning Traces with Legal Issue Trees. *ACL 2026 Main Conference (Oral)*.
- [C6] Seungone Kim, Ian Wu*, **Jinu Lee***, Xiang Yue, Seongyun Lee, Mingyeong Moon, Kiril Gashteovski, Caroline Lawrence, Julia Hockenmaier, Graham Neubig, Sean Welleck (2026). Scaling Evaluation-time Compute with Reasoning Models as Process Evaluators. *ACL 2026 Findings*.
- [C5] **Jinu Lee**, Julia Hockenmaier (2025). Evaluating Step-by-step Reasoning Traces: A Survey. *EMNLP 2025 Findings*.
- [C4] Chaeun Kim, **Jinu Lee**, Wonseok Hwang (2025). LegalSearchLM: Rethinking Legal Case Retrieval as Legal Elements Generation. *EMNLP 2025 Main Conference*.
- [C3] **Jinu Lee**, Qi Liu, Runzhi Ma, Vincent Han, Ziqi Wang, Heng Ji, Julia Hockenmaier (2025). Entailment-Preserving First-Order Logic Representations for Natural Language Entailment. *ACL 2025 Main Conference*.
- [C2] **Jinu Lee**, Wonseok Hwang (2025). SymBa: Symbolic Backward Chaining for Structured Natural Language Reasoning. *NAACL 2025 Main Conference*.
- [C1] Youngwon Lee*, **Jinu Lee***, Seung-won Hwang (2023). Learning to Rank Generation with Pairwise Partial Rewards. *EMNLP 2023 Main Conference*.

Preprints/Workshops.

- [P4] Jaeyong Ko, **Jinu Lee**, Pilsung Kang, Yukyung Lee (2026). Cliff Tokens: Identifying Single-Token Failure Triggers in LLM Mathematical Reasoning. *arXiv preprint arXiv:2606.25524*
- [P3] **Jinu Lee**, Shraddha Barke (2026). Symbolic Constraint Agents: Mining Tool-Call Constraints from Documentation and Traces. *Who Verifies the Agents? Workshop @ NeurIPS 2026..*
- [P2] Amruta Parulekar, **Jinu Lee**, Dilek Hakkani-Tur, Hari Sundaram (2025). Reasoning Consensus: Structural Ensembling of LLM Reasoning via Weighted DAG Aggregation. *arXiv preprint arXiv:2607.27783*.
- [P1] Priyanka Kargupta*, Stella Li*, Haocheng Wang, **Jinu Lee**, Shan Chen, Orevaoghene Ahia, Dean Light, Thomas L. Griffiths, Max Kleiman-Weiner, Jiawei Han, Asli Celikyilmaz, Yulia Tsvetkov (2025). Cognitive Foundations for Reasoning and Their Manifestation in LLMs. *HC-AIR workshop @ ICLR 2026*.

*: equal contribution.

TEACHING, TALKS, AND SERVICE

Teaching

CS307 Modeling and Learning in Data Science

Teaching Assistant

Urbana, IL

01/2026-05/2026

CS440 Artificial Intelligence

Teaching Assistant

Urbana, IL

08/2025-12/2025

Talks

ReasoningFlow: Semantic Structure of Complex Reasoning Traces

EleutherAI

Virtual

01/2026

Evaluating Legal Reasoning Traces with Legal Issue Trees

IDA.L Lab Seminar @ UoS

Seoul, Rep. of Korea

01/2026

Coupling Formal Logic and Natural Language Reasoning

Human-AI Collaborative Coding Lab @ Yonsei/KNU/UoS/SKKU

Seoul, Rep. of Korea

02/2025

Academic Service

ACL Rolling Review (ARR) Reviewer

10/2024-Present

Conference on Language Modeling (CoLM) Reviewer

03/2025-Present

Midwest Speech & Language Day (MSLD) 2026 Student organizer

04/2026

NeurIPS Evaluation & Datasets Track Reviewer

05/2026-Present

AWARDS AND HONOURS

AICE Ph.D. Fellow

Awarded by Amazon-Illinois Center on AI for Interactive Conversational Experiences

08/2026 - 05/2027

KFAS Overseas Ph.D. Scholarship

Awarded by Korea Foundation of Advanced Studies

08/2024 - 05/2029

Presidential Science Scholarship for Undergraduates

Awarded by Korea Student Aid Foundation

03/2018 - 08/2023